FastAPI Microservice Specification - Version 1.2

Section 1: Server Configuration and Architecture

DevDocs AI microservices architecture relies on FastAPI running on top of Uvicorn ASGI server.

The core authentication and routing service binds by default to HTTP port 8000.

The primary service health check endpoint is exposed at GET /api/v1/health returning status ok.

Worker concurrency is governed by Uvicorn process workers configured via WEB_CONCURRENCY environment variable.

Authentication in Version 1.2 is handled through standard HTTP Bearer Tokens.

Clients must obtain a token via POST /api/v1/auth/login with valid client credentials in JSON format.

Tokens are signed with HMAC-SHA256 and carry a validity TTL of 3600 seconds (1 hour).

All internal service communications use unencrypted HTTP over the local Docker bridge network.

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Section 2: Rate Limiting and Resilience Policy

In Version 1.2, rate limiting is applied at the API gateway layer per client IP address.
The standard throughput threshold is fixed at 100 requests per minute per remote IP address.
When a client exceeds this limit, the gateway immediately returns HTTP 429 Too Many Requests.
A Retry-After header with a fixed delay of 60 seconds is sent with every 429 response.
For persistent data storage, the service connects to MongoDB Atlas using standard connection strings.
Connection pool size is capped at 50 connections with a server selection timeout of 5000 milliseconds.
Application logs are emitted in JSON format to stdout for log collection by fluentd.